Session 13 - Basics of RMarkdown

This is my subtitle

Member A Member B Member C Member D

2025-12-08

Text Editing in Markdown

- This is how we add links in R Markdown:

You simply write as if it were a regular text editing document.

This is an R Markdown document. Markdown is a formatting syntax tool for creating HTML, PDF, and MS Word documents. For more details on how to use R Markdown, see http://rmarkdown.rstudio.com.

- We can also add hyperlinks:

This is an example of how to add hyperlinks. You can read more about R Markdown at this link.

- This is how we specify titles in RMarkdown:

Title 1 (highest hierarchy)

Above was the highest hierarchy

Title 2 (second-level hierarchy)

Above was the second highest hierarchy

Title 3 (third-level hierarchy)

- We can edit our text by adding formatting:

This is an example of **bold** text. I can also **use it in full sentences or multiple words**.

We can format words in **bold**, *italics* or *underlined*

> We can include a paragraph with more indentation than a regular paragraph.

We can also include a footnote anywhere in the document.[1]

[1] Footnote text. Here we can also embed links that redirect to websites.

Building a Numbered List

1. Item 1
2. Item 2
3. Test item

Building a Bullet List

- Item 1
- Item 2
- Test item 3
- Test item 4

Bullet List with a Nested Numbered Sublist (4 spaces indentation)

- Item 1
  1.1. This is my first sub-element
- Item 2
  1. Sub-element 1
  2. Sub-element 2
  3. Another sub-element
- In R Markdown we can also add LaTeX equations:
  Inline equation example: $\alpha = \beta_0 + \beta_1 \times coefficient + \lambda^2$.

Centered equation example:

$$\alpha = \beta_0 + \beta_1 \times coefficient + \frac{1}{2}$$

This link provides a useful guide for operators, symbols, and Greek letters.

Tambien podemos agregar las ecuaciones centradas (agregando dos simbolos de dolar)

$$\alpha = \beta_0 + \beta_1 \times coefficient$$

Este enlace es una guia util para distintos operadores, símbolos y alfabeto griego para que puedan hacer sus ecuaciones.

Tables in R Markdown

We often want to add tables from data frames into our reports. We can add a code chunk for this:

```
## # A tibble: 4 x 4
##   Region        Average homicide rat~1 `Number of states` Total population by ~2
##   <fct>                          <dbl>              <int>                  <dbl>
## 1 Northeast                       346.                  9               55317240
## 2 South                           431.                 17              115674434
## 3 North Central                   244.                 12               66927001
## 4 West                            693.                 13               71945553
## # i abbreviated names: 1: `Average homicide rate`,
## #   2: `Total population by region`
```

Table 1: Firearm homicides in the U.S. for 2010

Region	Average homicide rate	Number of states	Total population by region
Northeast	346.0922	9	55317240
South	430.8356	17	115674434
North Central	243.7860	12	66927001
West	693.3508	13	71945553

We can do exactly the same with regression!

```
## OLS estimation, Dep. Var.: childHeight
## Observations: 934
## Standard-errors: IID
##                  Estimate Std. Error  t value   Pr(>|t|)
## (Intercept)     22.636241   4.265107  5.30731 1.3909e-07 ***
## midparentHeight  0.637361   0.061608 10.34549  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 3.38808   Adj. R2: 0.102046
```

Table 2: Child height as a function of parent height

	(1)
Intercept	22.636 ***
	(4.265)
Average parent height (inches)	0.637 ***
	(0.062)
N. obs.	934
R squared	0.103

* p < 0.1; ** p < 0.05; *** p < 0.01.

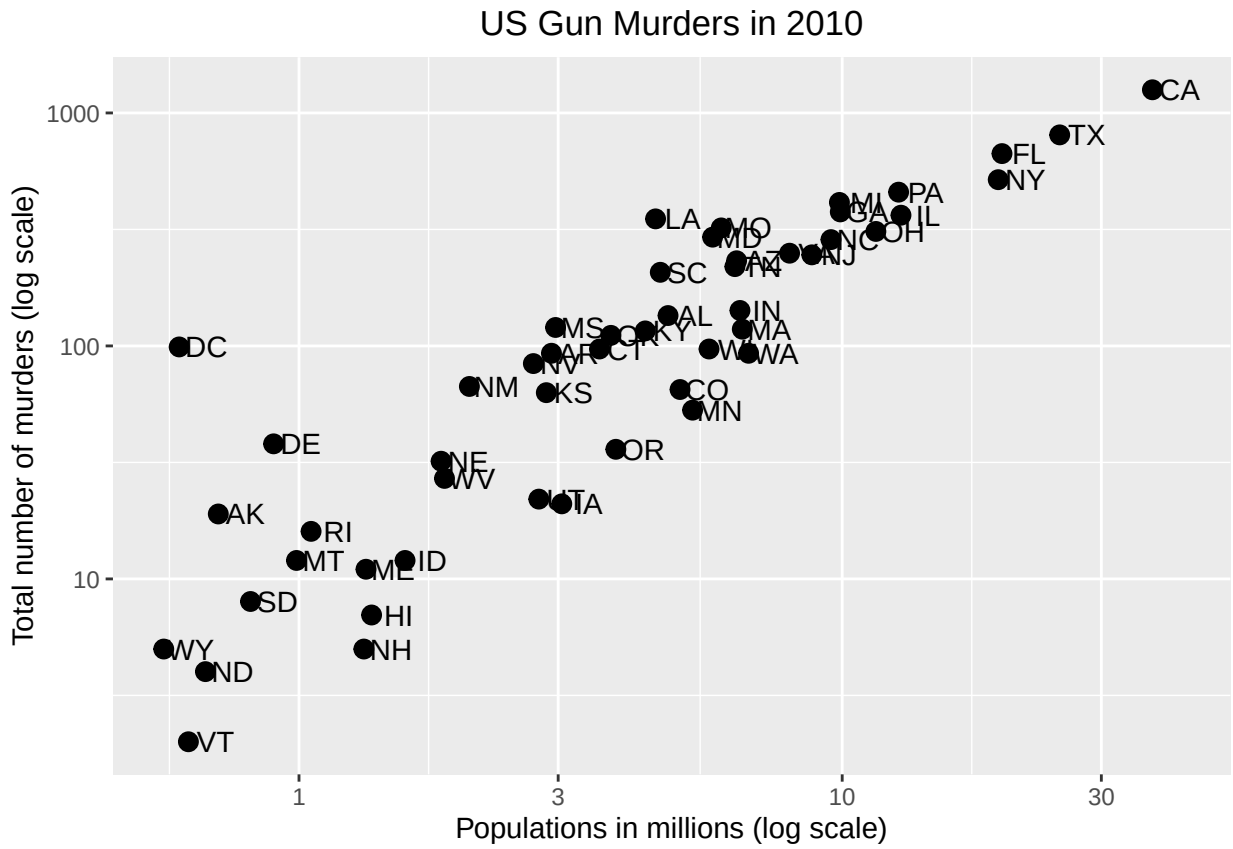
Table 3: Child height as a function of parent height

	(1)	(2)
Intercept	22.636 ***	32.023 ***
	(4.265)	(3.311)
Average parent height (inches)	0.637 ***	0.549 ***
	(0.062)	(0.048)
Number of children in household		-0.917 ***
		(0.036)
N. obs.	934	934
R squared	0.103	0.467

* p < 0.1; ** p < 0.05; *** p < 0.01.

Add graphs into R Markdown

Just as we embedded tables, we can embed plots:



Referencias en R Markdown

With what we've learned so far, we can build a simple reference section at the end of our reports. But there is a more powerful option:

There are reference managers like Zotero allow us to store references online (or via DOI). This is extremely helpful for academic papers or theses, because Zotero can automatically generate a bibliography section. Zotero is compatible with R Markdown, MS Word, and Google Docs.

Esto resulta particularmente útil para trabajos universitarios o trabajos de grado, debido a que haciendo uso de Zotero correctamente, nos permite generar de forma automática la sección de referencias al final de nuestro informe o trabajo. Zotero es compatible con R Markdown, así como es compatible con Microsoft Word y Google Docs.

I recommend exploring Zotero for adding citations (especially thesis work).

Here is also a useful a link explaining how to use Zotero with R Markdown through LaTeX bibliography formatting.